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Wildfire Susceptibility Mapping Using Deep Learning and Machine Learning Models Based on Multi-Sensor Satellite Data Fusion: A Case Study of Serbia

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Abstract

To prevent or mitigate the negative impact of fires, spatial prediction maps of wildfires are created to identify susceptible locations and key factors that influence the occurrence of fires. This study uses artificial intelligence models, specifically machine learning (XGBoost) and deep learning (Kolmogorov-Arnold networks—KANs, and deep neural network—DNN), with data obtained from multi-sensor satellite imagery (MODIS, VIIRS, Sentinel-2, Landsat 8/9) for spatial modeling wildfires in Serbia (88,361 km^2). Based on geographic information systems (GIS) and 199,598 wildfire samples, 16 quantitative variables (geomorphological, climatological, hydrological, vegetational, and anthropogenic) are presented, together with 3 synthesis maps and an integrated susceptibility map of the 3 applied models. The results show a varying percentage of Serbia's very high vulnerability to wildfires (XGBoost = 11.5%; KAN = 14.8%; DNN = 15.2%; Ensemble = 12.7%). Among the applied models, the DNN achieved the highest predictive performance (Accuracy = 83.4%, ROC-AUC = 92.3%), followed by XGBoost and KANs, both of which also demonstrated strong predictive accuracy (ROC-AUC > 90%). These results confirm the robustness of deep and machine learning approaches for wildfire susceptibility mapping in Serbia. SHAP analysis determined that the most influential factors are elevation, air temperature, and humidity regime (precipitation, aridity, and series of consecutive dry/wet days).

Keywords: artificial intelligence (AI); Kolmogorov-Arnold networks (KANs); deep neural network (DNN); extreme gradient boosting (XGBoost); Shapley additive explanations (SHAP); Sentinel-2; Landsat 9; environmental protection; geographic information systems (GIS); remote sensing

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1. Introduction

Wildfires are a natural disaster, defined as an unplanned fire caused by humans or specific natural conditions [1]. In natural settings, fires are typically formed due to the spontaneous combustion of extremely dry fuel material, often caused by an intense heat wave or prolonged dry period [2]. In exceptional cases, fires can also occur due to lightning strikes, but the human factor is the predominant factor in initiating fires, which is responsible for 89% of wildfires in the world [3]. Large-scale fires cause negative consequences for the environment, ecosystems, infrastructure, agricultural plots, and local industry [4]. They affect the reduction in air quality, changes in soil structure, and pose a real threat to the local population and their property [5].

Due to the intensification of climate extremes (extended dry seasons, high temperatures) and human activities, wildfires are projected to increase in the coming years [6]. In Germany, between 400 and 3000 fires occur each year, covering an area of almost 5000 hectares [5]. In Russia, economic losses from wildfires amount to 0.06% of GDP per year, with approximately 800,000 hectares of forest and timber damaged [7]. In Australia, fires destroyed more than 5900 structures and 18.6 million hectares of land, and 34 people lost their lives during the 2019–2020 season [6]. In North America, the 2018 Camp Fire in California claimed 85 lives and destroyed approximately 19,000 buildings [6]. On the other hand, the 2023 Maui fires resulted in almost a hundred deaths and an estimated USD 550 million in property damage [8]. Between 2011 and 2020, more than 30,000 wildfires were recorded in China, affecting a total area of 155,700 hectares and resulting in 543 casualties [9]. Catastrophic fires in Serbia occurred on 7 July 2025. On that day, more than 600 fires were registered in Serbia, particularly affecting the central part (Šumadija region) and the southern part (Toplički district) of the country. It is estimated that the fires destroyed more than 300 houses, damaged thousands of hectares of agricultural and forest land, and injured several firefighters and citizens [10].

One of the most important preventive measures for effective wildfire management is the development of maps of the spatial distribution of wildfires [11]. Fire susceptibility prediction models involve the processing and analysis of a large dataset on natural conditions (topography, climatology, vegetation, hydrology), socio-economic variables, and the history of previous fires [12]. With this approach, it is possible to achieve precise zoning of endangered areas and minimize the environmental impact [13]. In modern research, qualitative, semi-quantitative, and quantitative approaches are employed to predict fires and other natural disasters, phenomena, and environmental processes [14,15]. Qualitative and semi-quantitative methods were first developed that involved the use of expert judgment and hierarchical analysis in their analyses [16]. In such cases, the results depend largely on the subjective assessment of experts, which limits the accuracy of the results. In contrast, machine learning (ML) models offer a modern quantitative approach based on a large dataset of physical-geographic and socio-economic variables, as well as a wildfire inventory formed on the basis of historical data [17]. Numerous studies have shown that ML methods achieve the highest accuracy when large and relevant datasets are available [7]. In Greece, Symeonidis et al. (2025) used four ML models (XGBoost, GBM, LightGBM, and CatBoost) to estimate wildfire susceptibility [18]. Moghim and Mehrabi applied ML methods (logistic regression and random forest) to predict wildfires in parts of the United States and Canada in 2024 [19]. Ismail et al. (2025) used Variational AutoEncoder, Support Vector Machine, AutoEncoder, Isolation Forest, Deep Support Vector Data Description, and Adversarially Learned Anomaly Detection to spatially predict wildfires in California and Western Australia [20]. Milanović et al. (2021) applied the Random Forest (RF) and Logistic Regression (LR) models to map forest fire probability in Eastern Serbia [21].

As a separate branch of machine learning, deep learning approaches have been actively applied in scientific practice in recent years. In combination with satellite imagery, i.e., multi-sensor data fusion, deep learning models often show more accurate performance for wildfire prediction than classical ML methods [6]. Bahadori et al. (2023) used deep learning (recurrent neural network and long short-term memory) with MODIS and Landsat-8 satellite imagery for the area of western Iran affected by fires in 2021 to generate wildfire susceptibility maps [22]. Bjånes et al. (2021) applied an Ensemble model based on deep neural networks (CNNs and other architectures), along with machine learning algorithms XGBoost and SVM, to generate wildfire susceptibility maps for two regions in Chile (2013–2019) using satellite imagery and 15 wildfire-influencing factors [23].

He et al. (2024) employed an optimized ConvLSTM model enhanced with channel and spatial attention mechanisms and a Vision Transformer module, applied to multi-source satellite, meteorological, vegetation, and anthropogenic data across six eastern provinces of China (2012–2022) [24]. Jiang et al. (2024) developed a CNN-based wildfire risk assessment model trained on 11,507 wildfire cases in Guangdong Province (2011–2021), integrating four categories of driving factors (topography, vegetation, weather, and human activities), and demonstrated that their model outperformed traditional machine learning methods (DT, RF, Logistic, KNN, SVM) [9]. For all applied approaches, cartographic interpretation of input parameters and final results is important. In this case, geographic information systems (GIS) and remote sensing data play a key role, facilitating and more clearly depicting the susceptibility of terrain to fires through space and time [24–30].

Our study employs three deep and machine learning models: Deep Neural Network (DNN), Kolmogorov–Arnold Networks (KANs), and eXtreme Gradient Boosting (XGBoost) with the aim of spatially predicting wildfire vulnerability in Serbia's territory using 16 predictive variables based on multi-sensor satellite data. All predictors represent long-term environmental, topographic, and climatic characteristics derived as multi-year averages (1991–2020), ensuring spatial consistency across the study area. Consequently, the models estimate the spatial susceptibility of wildfire occurrence rather than short-term temporal dynamics (e.g., year-to-year prediction).

The main objectives of the study include the following:

1. Creation of a spatial database with almost 200,000 fires and indicators with 16 quantitative variables;
2. Generating spatial wildfire hazard maps at the national level;
3. Integrating spatial data from all models and creating a synthesis hazard map;
4. Comparative analysis of the performance of deep and machine learning models;
5. Discovering key factors contributing to fire occurrence based on SHAP analysis;
6. Identification of the most susceptible municipalities in Serbia to wildfires.

The applied models, their integration, and the obtained results will serve as an effective tool for discovering the spatial patterns of wildfires and their distribution. This is the first study to utilize the DNN and KANs models for national-level spatial wildfire prediction.

2. Materials and Methods

2.1. Study Area

Serbia is one of the largest countries in Southeast Europe. It is located in the central part of the Balkan Peninsula, at an important geographical and transport crossroads that connects the Pannonian Plain with the Carpathian-Balkan Mountains, then Central Europe with the Aegean and Adriatic Seas [31].

The total area is 88,361 km^2, and the largest cities are Belgrade (the capital), Novi Sad, Niš, and Pristina (Figure 1). According to the 2022 census, the population is 6.647 million (without data for the Autonomous Province of Kosovo and Metohija) [32]. The geographical position of Serbia has significantly influenced the diversity of natural conditions.

Geomorphologically, the relief of Serbia can be divided into three main units: the plain in the north, the hilly and mountainous central part, and the mountainous regions. In the north lies the Pannonian Plain, which covers most of Vojvodina and is characterized by fertile alluvial soils, suitable for intensive agricultural production. The central part of Serbia is characterized by the predominantly hilly and mountainous region of Šumadija [31]. At the same time, the south includes mountain ranges such as Kopaonik, Stara Planina, Prokletije, and Šar Planina. The altitude of Serbia ranges from 28 m (the confluence of the Timok and the Danube) to 2660 m (the peak of Velika Rudoka on Šar Planina) [33].

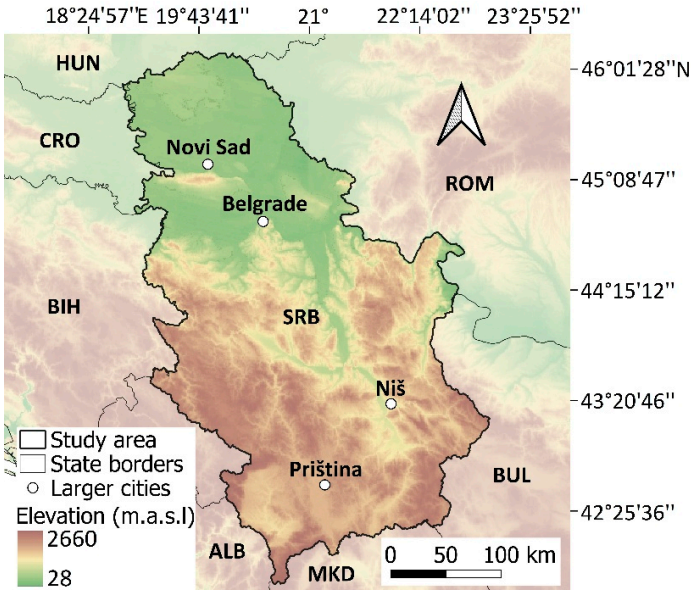


Figure 1. Geographical position of Serbia.

The rivers of Serbia belong to the largest European basins, primarily the Danube basin. The Danube flows through the northern part of the country for approximately 588 km, serving as a significant water corridor. Other major rivers are the Sava, Tisa, Drina, Ibar, and Velika Morava [34]. The climate of Serbia is predominantly temperate continental, characterized by distinct seasons. Winters are cold with frequent frosts and snow, while summers are warm and dry. In mountainous areas, the climate is harsher and subalpine, whereas in the Pannonian Plain, there are influences of a continental climate with larger temperature fluctuations [31].

Biodiversity and natural protected areas are particularly pronounced due to the combination of diverse climatic, geomorphological, and hydrological conditions. National parks (Đerdap, Fruška Gora, Tara, Kopaonik, Šar Planina) and special nature reserves represent significant areas for the preservation of flora and fauna.

Due to the very rich biodiversity and high-quality productive land in Vojvodina, it is necessary to create maps with the spatial susceptibility of wildfires in order to implement effective protection measures (Figure 2). Spatial prediction of wildfires in Serbia is conducted due to the increasing frequency and destructiveness of such events, as exemplified by the large-scale wildfire on 7 July, which caused extensive environmental damage, endangered human settlements, and highlighted the urgent need for advanced risk assessment and prevention strategies [10].



Figure 2. Wildfires in Serbia, and their impact on buildings (**a**,**c**) and forests (**b**,**d**) [35–38].

2.2. Dataset Construction

2.2.1. Historical Wildfires and Inventory

For monitoring wildfires, satellite observations are a primary tool, as they enable coverage of large areas and provide significant information on the spatiotemporal patterns of fires. Fire data were obtained from NASA's Fire Information for Resource Management System (FIRMS) [39].

Historical records for the period from 2001 to 2024 were analyzed, using data from the VIIRS (Visible Infrared Imaging Radiometer Suite) and MODIS (Moderate Resolution Imaging Spectroradiometer) instruments. The spatial resolution of VIIRS data is 375 m, while the MODIS pixel size is 1000 m [39]. A total of 199,598 fires were registered in Serbia's territory over 24 years, as documented in the created inventory database [39]. Data on fires marked as nominal (N) and high (H) confidence values were used in the analysis, while data with low confidence were excluded. The data were then processed and displayed using QGIS 3.28.10 software [40]. Each point in the database contains information about the location and time of the fire (Figure 3).

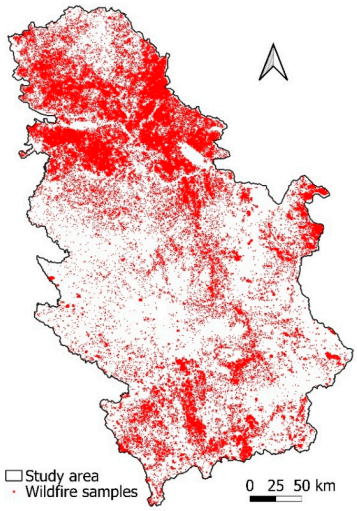


Figure 3. Wildfire inventory map in Serbia for the period 2001–2024.

2.2.2. Topographic Characteristics

Topographic (geomorphological) characteristics play a crucial role in assessing wildfire susceptibility. The topographic variables were derived from the European Digital Elevation Model (EU-DEM), developed by the European Environment Agency [41]. With a spatial resolution of 25 m, the EU-DEM provides highly accurate elevation data. Slope, aspect, and topographic wetness index were calculated using the same dataset, while all spatial analyses were performed in QGIS 3.28.10 [40].

The elevation within the study area ranges between 28 and 2660 m, resulting in a total vertical difference of 2632 m (Figure 4). Topography influences wildfire occurrence by 185 shaping climatic and vegetation conditions [42]. As altitude increases, air temperature decreases while relative humidity rises, leading to a higher frequency of wildfires at lower elevations [43]. Terrain slope influences wildfire dynamics by controlling both fuel availability and the rate of fire spread.

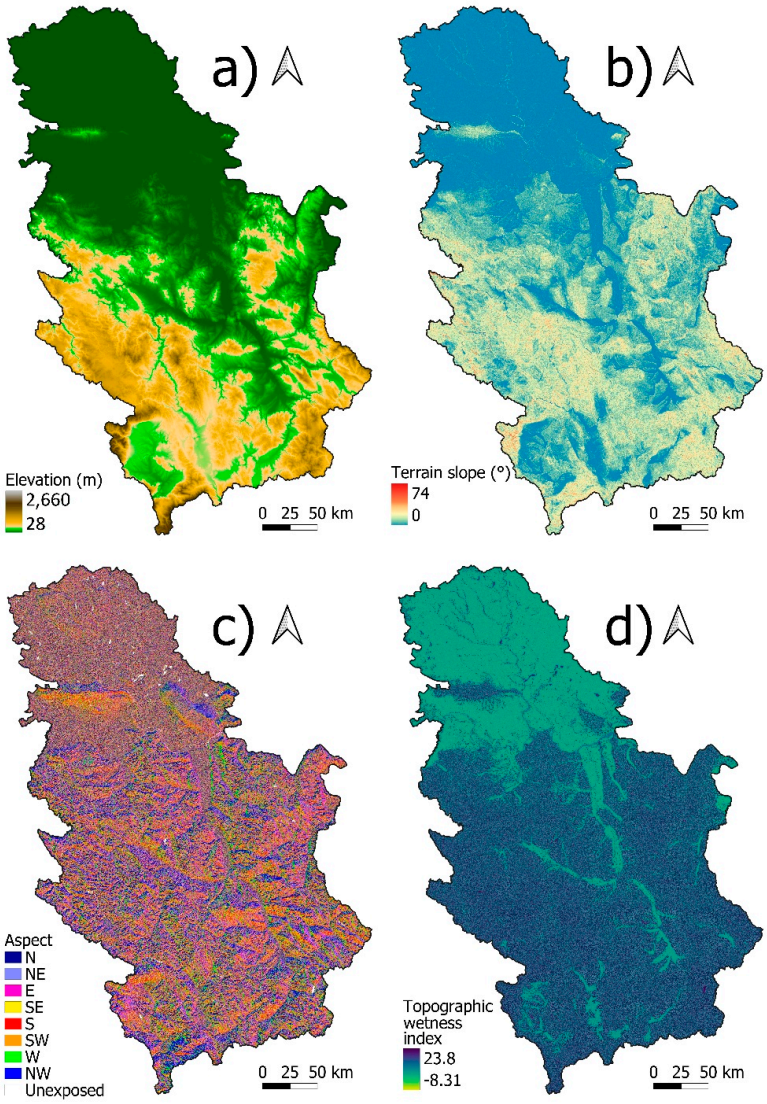


Figure 4. Topography conditions: (**a**) elevation; (**b**) slope; (**c**) aspect; (**d**) topographic wetness index.

Fires tend to propagate more rapidly upslope because flames are tilted closer to the surface, preheating and igniting vegetation more efficiently [42]. Aspect affects fuel dryness: south-facing slopes in the Northern Hemisphere receive more sunlight, leading to lower fuel moisture and higher fire susceptibility, while shaded slopes (north-facing) tend to retain more humidity [43].

The Topographic Wetness Index (TWI) indicates areas where water tends to accumulate. Locations with low TWI values are drier and therefore more prone to ignition. In contrast, higher TWI areas usually have greater soil moisture and denser vegetation, which reduces fire probability but potentially provides more biomass if they do ignite [17].

2.2.3. Climate Conditions

For the climatological analysis, the dataset included seven variables: mean air temperature, total annual precipitation, number of consecutive dry days (CDD), number of consecutive wet days (CWD), aridity index, global horizontal irradiance, and wind exposure. Information on temperature, precipitation, CDD, CWD, and aridity index was obtained from the Digital Climate Atlas of Serbia, which is based on measurements from climatological and meteorological stations for the period 1981–2010 [44]. Average values are shown for the observed 30-year period. These data were interpolated and resampled into a GIS environment with a spatial resolution of 25 m (Figure 5).

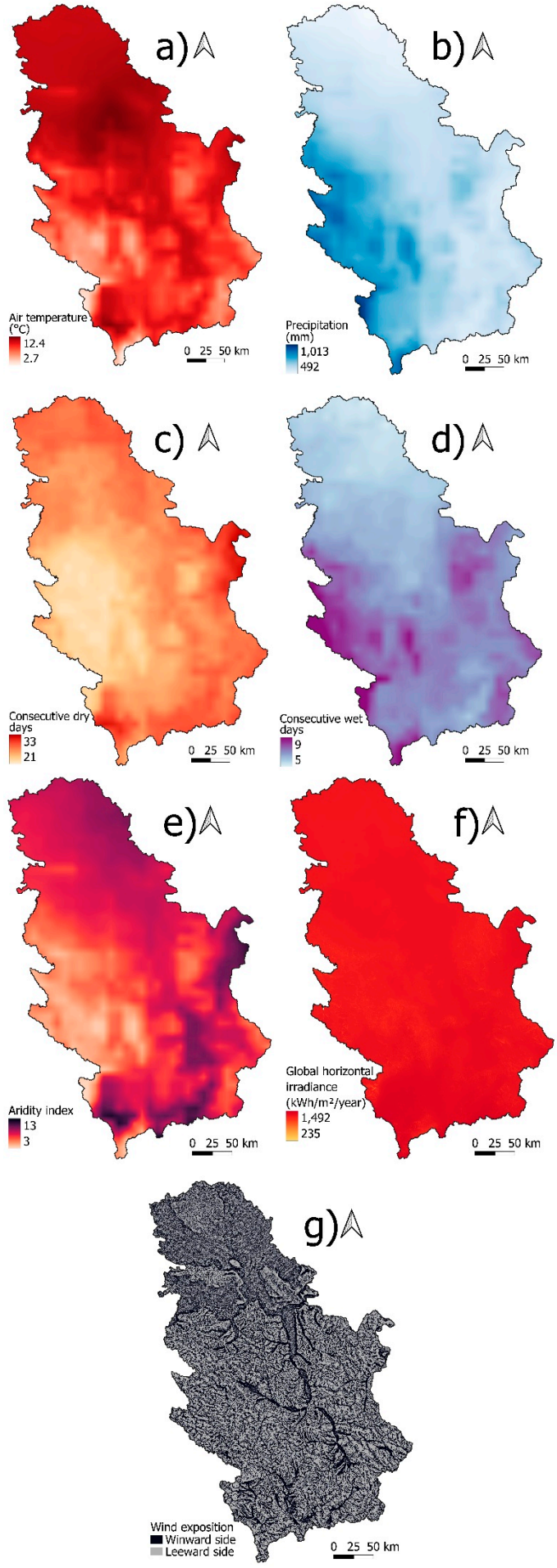


Figure 5. Climate conditions: (**a**) air temperature; (**b**) precipitation; (**c**) consecutive dry days; (**d**) consecutive wet days; (**e**) aridity index; (**f**) global horizontal irradiance; (**g**) wind exposition.

Global horizontal irradiance for the territory of Serbia was obtained by processing data from the Global Solar Atlas [45]. Wind exposure was derived from the EU-DEM using QGIS tools, where the windward and leeward slopes were distinguished to produce the final dataset [40]. In the SAGA plugin Wind Exposition Index, the EU-DEM was imported, after which the wind exposure was determined. The spatial resolution of this layer is 25 m. Wildfire occurrence and behavior are strongly conditioned by climate. Higher mean air temperatures and lower annual precipitation reduce fuel moisture, while prolonged periods of consecutive dry days (CDD) further increase the likelihood of ignition [29]. Conversely, consecutive wet days (CWD) contribute to fuel accumulation but may also delay drying, indirectly influencing fire risk [46]. The aridity index integrates the balance between precipitation and potential evapotranspiration, providing a measure of long-term dryness that governs vegetation stress and flammability [12]. In addition, global horizontal irradiance accelerates surface drying through increased solar energy input, whereas wind exposure affects both the drying rate of fuels and the speed and direction of fire spread [47].

2.2.4. Hydrological Characteristics

The most significant hydrological factor in spatial prediction of wildfires is the distance from water surfaces. Data on water bodies were obtained from the Environmental Systems Research Institute platform [48]. From the water bodies, all major rivers, natural and artificial lakes, as well as canals were selected (Figure 6).

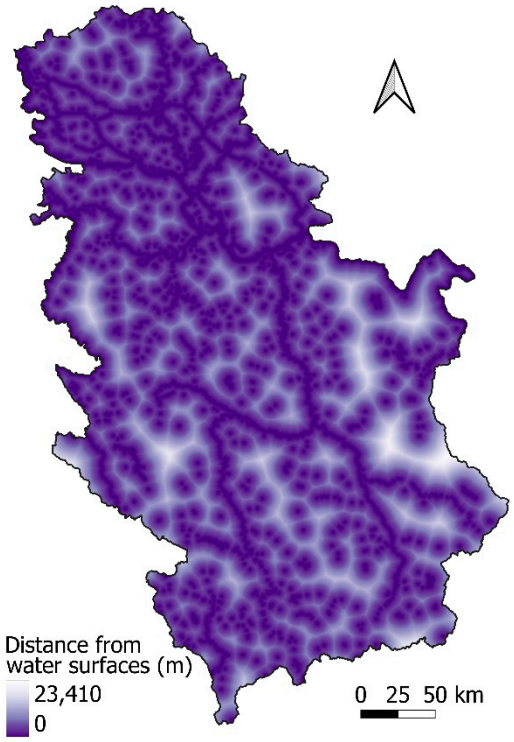


Figure 6. Hydrological condition: distance from water surfaces.

2.2.5. Vegetation Characteristics

Land use information was obtained from the Sentinel-2 Land Cover Explorer platform, which relies on satellite imagery from 2024 [48]. The vegetation across Serbia is categorized into eight land cover classes: water bodies, forests, bare soil, flooded zones, agricultural land, settlements, snow and ice, as well as meadows and pastures (Figure 7). Among these, forests, agricultural lands, meadows and pastures are particularly prone to wildfires due to the abundance of dry biomass that can serve as fuel [12]. Pixels have a spatial resolution of 10 m.

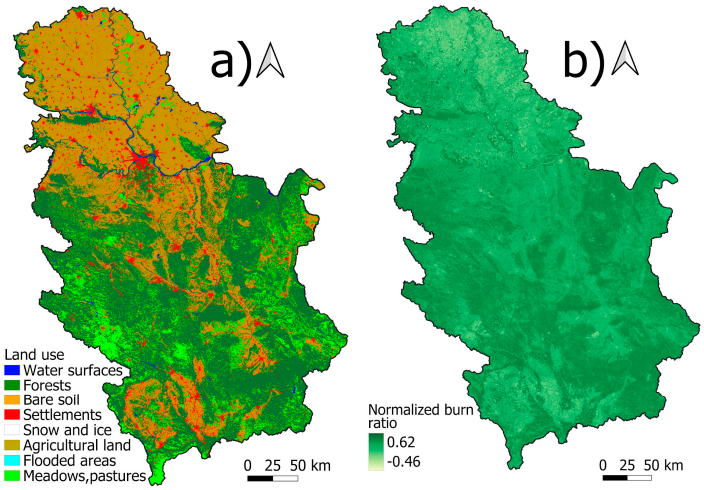


Figure 7. Vegetation characteristics: (**a**) land use; (**b**) normalized burn ratio.

The Normalized Burn Ratio (NBR) was obtained by merging multiple satellite images of Serbia's territory from July 2025, with cloudiness less than 0.2%. For this purpose, Landsat 8 and Landsat 9 satellite sensors were used, whose data have a spatial resolution of 30 m [49]. Satellite scenes from 2025 were used as they provide a relevant analysis of the current state of vegetation, particularly since they were captured after the large fires in July. *NBR* is obtained based on the formula [50]:

$$NBR = \frac{(NIR - SWIR_2)}{(NIR + SWIR_2)}, \quad (1)$$

where *NIR* is the near-infrared spectral band, and $SWIR_2$ is the short-wave infrared spectral channel. High values indicate healthy vegetation, low values indicate recently burned areas, while values close to zero indicate unburned areas [50].

2.2.6. Anthropogenic Factors

Among the anthropogenic variables, two were considered: distance from roads and distance from settlements (Figure 8). Settlements were extracted from the land use dataset (Environmental Systems Research Institute), after which the distance from each pixel to the nearest settlement was computed in QGIS [40,48]. For roads, features were digitized and downloaded from the OpenStreetMap platform [51].

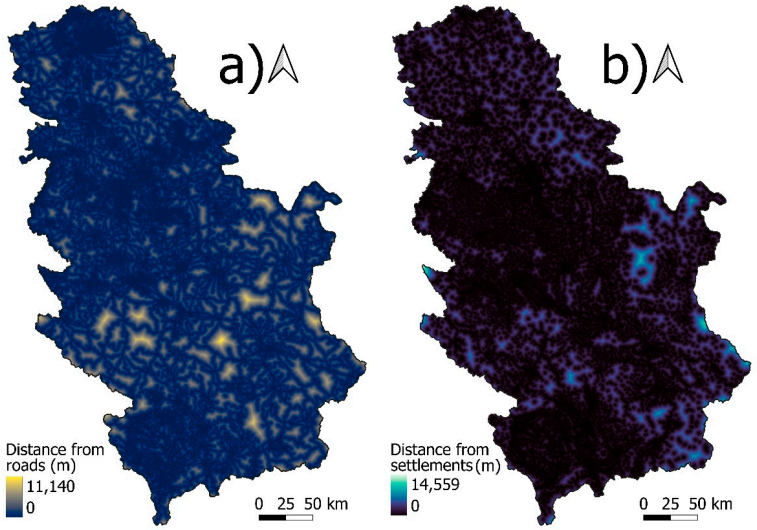


Figure 8. Anthropogenic factors: (**a**) distance from roads; (**b**) distance from settlements.

All measurements were expressed in meters, with an initial spatial resolution of 25 m. To ensure comparability and spatial consistency, the resolution of all 16 predictive variables was resampled to 25 m. Proximity to roads and settlements increases the likelihood of wildfires due to greater human influence and possible ignition sources. In contrast, proximity to water bodies generally acts as a natural barrier, reducing the risk of fire occurrence and spread [13].

Table 1 presents in detail the analyzed natural and anthropogenic conditions, their spatial resolution, and data sources. Most predictors represent long-term spatial averages or multi-year summaries (1991–2020) rather than annual time series. Consequently, the resulting dataset describes stable environmental conditions rather than year-specific variability.

Table 1. Quantitative variables and dataset sources.

Variable	Year	Spatial Resolution (m)	Source
Elevation	2016	25	European Environment Agency
Terrain slope	2016	25	European Environment Agency
Aspect	2016	25	European Environment Agency
Topographic wetness index	2016	25	European Environment Agency
Air temperature	2022	25 (resampled)	Digital Climate Atlas of Serbia
Annual precipitation	2022	25 (resampled)	Digital Climate Atlas of Serbia
Consecutive dry days	2022	25 (resampled)	Digital Climate Atlas of Serbia
Consecutive wet days	2022	25 (resampled)	Digital Climate Atlas of Serbia
Aridity index	2022	25 (resampled)	Digital Climate Atlas of Serbia
Global horizontal irradiance	2025	25 (resampled)	Global Solar Atlas
Wind exposure	2016	25 (resampled)	European Environment Agency
Land use	2024	25 (resampled)	Environmental Systems Research Institute
Normalized Burn Ratio	2025	25 (resampled)	U.S. Geological Survey
Distance from roads	2025	25	Open Street Map
Distance from settlements	2025	25	Environmental Systems Research Institute
Distance from water surfaces	2025	25	Environmental Systems Research Institute

2.3. Methodology

Before model development, it was necessary to establish a consistent and representative dataset that integrates all spatial predictors and the wildfire inventory. This ensured that each modeling approach would operate on identical inputs, enabling a fair comparison of their predictive capabilities.

2.3.1. Deep Learning Techniques and Network Models

We first constructed a unified, analysis-ready dataset from the 16 geomorphological, climatological, hydrological, vegetational, and anthropogenic predictors and the wildfire inventory. All raster predictors were harmonized to the same coordinate reference system, spatial extent, and cell size. Continuous layers (e.g., temperature, precipitation, irradiance, elevation, slope, TWI, NBR, distances) were resampled with bilinear interpolation to preserve gradients, whereas categorical layers (e.g., land use, aspect) used nearest-neighbor

to avoid class mixing. The wildfire inventory was rasterized onto this common grid and converted to a strict binary target (0/1), explicitly removing NoData so that non-fire pixels are coded as 0 everywhere in the domain.

From the harmonized stacks, we performed pixel-level sampling to assemble the modeling table. To limit redundancy and computational load while retaining spatial representativeness, we used systematic grid sampling (every third pixel in both axes). Categorical predictors were one-hot encoded into mutually exclusive indicator columns; cloud/invalid categories (when present in land use data) were excluded from emission and from sampling. Any pixel with NoData in any predictor was dropped to prevent leakage of missingness patterns. The target label was derived as 1 for any location with at least one recorded fire event in the reference grid cell and 0 otherwise. Because the raw distribution of fire versus no-fire is highly imbalanced, we applied class-aware sampling to achieve a controlled negative-to-positive ratio of approximately 1.5:1. This preserves all positive instances while subsampling negatives to the desired proportion, improving learning stability without excessively distorting the underlying prevalence.

After sampling, numeric features were min–max scaled to [0, 1]; one-hot (categorical) indicators remained as 0/1. Altogether, the 16 conceptual predictors expanded into 29 encoded input features, which served as the effective inputs to the models. Geographic coordinates were retained only as metadata for potential mapping and were not used as model inputs to avoid spatial leakage. Quality control checks ensured consistent array shapes, absence of residual missing values, stable feature ordering, and reproducible sampling via fixed random seeds. This unified dataset, balanced, normalized, and free of missing values, served as the common input for training the three model families evaluated in this study: a deep neural network (DNN), a Kolmogorov–Arnold network (KANs), and a gradient-boosted decision tree model (XGBoost). Each model consumed exactly the same feature matrix and label vector prepared by the workflow above.

2.3.2. Network Structure and Configuration

Three different model families were developed and configured to predict wildfire susceptibility: a deep neural network (DNN), a Kolmogorov–Arnold network (KANs), and an ensemble of decision trees (XGBoost). Each model was trained on the same dataset, allowing for a direct and fair comparison of their predictive capacity.

The DNN was built as a residual multilayer perceptron designed to capture complex nonlinear interactions among the 29 encoded input features derived from the 16 original predictors. Its structure consisted of several fully connected layers with progressively decreasing widths [52,53]. Each layer was followed by batch normalization to stabilize learning, SiLU (Sigmoid Linear Unit) activation functions to enhance nonlinear representation, and dropout regularization to reduce overfitting [54]. A residual connection scheme was incorporated to improve gradient flow and accelerate convergence. The model was trained using the AdamW optimizer with weight decay, dynamic learning rate scheduling based on the OneCycleLR policy, and automatic mixed-precision computations to leverage GPU efficiency [55]. Training was guided by early stopping to prevent unnecessary epochs once validation performance plateaued [56]. The DNN hyperparameters were empirically tuned within plausible ranges (learning rate 10^{-4}–10^{-3}, batch size 2048–8192, dropout 0.1–0.3, and weight decay 10^{-5}–10^{-3}). The final configuration (learning rate = 1×10^{-3}, dropout = 0.15, batch size = 4096) was selected based on the best validation F1-score, using a fixed random seed of 42 to ensure reproducibility.

The KANs employed a lighter residual MLP backbone but maintained the same principles of batch normalization, SiLU activation, and dropout [57]. Its configuration included fewer layers and narrower widths, making it computationally more efficient while still

able to approximate complex functional relationships among the 29 encoded input features derived from the 16 original predictors. A distinctive feature of this network was the use of an exponential moving average (EMA) of weights during training, which smoothed parameter updates and stabilized learning dynamics [58,59]. Like the DNN, it used AdamW optimization, OneCycleLR scheduling, gradient clipping, and early stopping criteria, ensuring robust performance even under class imbalance [60]. The KAN hyperparameters were tuned in the same search space as the DNN but with narrower layer widths (64–256 units). The final configuration (learning rate = 1×10^{-3}, dropout = 0.15, weight decay = 1×10^{-4}, EMA decay = 0.999) was selected as the most stable after multiple trials maximizing the validation F1-score. The random seed was again fixed to 42 for comparability across models.

The XGBoost model was implemented as a gradient-boosted decision tree ensemble designed to capture complex nonlinear relationships among the 29 encoded input features derived from the 16 original predictors. Configured for binary classification, it combined multiple shallow trees trained sequentially to minimize classification error [61]. Its configuration included a maximum depth of 8 for the trees, a learning rate of 0.05, and both row and column subsampling rates of 0.8 to reduce overfitting and improve generalization [62]. Class imbalance was corrected by adjusting the positive-to-negative scale weight based on the training distribution. The training process was monitored using logloss (logarithmic loss, also called logistic loss or cross-entropy loss), with early stopping applied after a predefined number of boosting rounds without improvement [63]. When available, GPU acceleration was used to speed up training, and gain-based feature importance scores were extracted from the fitted ensemble. For XGBoost, hyperparameters were optimized via grid search within the following ranges: max_depth = 4–10, learning_rate = 0.01–0.1, subsample = 0.6–1.0, colsample_bytree = 0.6–1.0, gamma = 0–0.3, and scale_pos_weight adjusted according to the class imbalance ratio. The final configuration (max_depth = 8, learning_rate = 0.05, subsample = 0.8, colsample_bytree = 0.8) was chosen based on the lowest validation logloss and highest F1-score, with random seed fixed to 42 for reproducibility.

All procedures and approaches employed in this research are outlined in the flow chart provided in Figure 9. The 16 conceptual predictors expanded into 29 encoded input features, which served as the effective inputs to the models.

The dataset was constructed to represent spatial variability across the study area rather than temporal sequences. Therefore, the models estimate spatial wildfire susceptibility, indicating areas more prone to fire occurrence under typical long-term conditions. Static predictors (e.g., topography, distances) remain constant, while dynamic variables (e.g., NDVI, NDMI, precipitation, temperature) were expressed as multi-year averages to reduce interannual noise.

2.4. Validation Strategy

Model performance was evaluated using spatial cross-validation (CV) with five spatially disjoint folds to reduce spatial autocorrelation. For each model, ROC and precision–recall (PR) curves were computed per fold, and performance metrics (AUC, F1-score, accuracy, precision, recall) were averaged across folds. Ninety-five percent confidence intervals for these metrics were estimated by bootstrap resampling (n = 1000) over folds. In addition, predictive uncertainty maps were generated by calculating the variance of susceptibility probabilities across folds, highlighting areas of greater model disagreement and uncertainty.

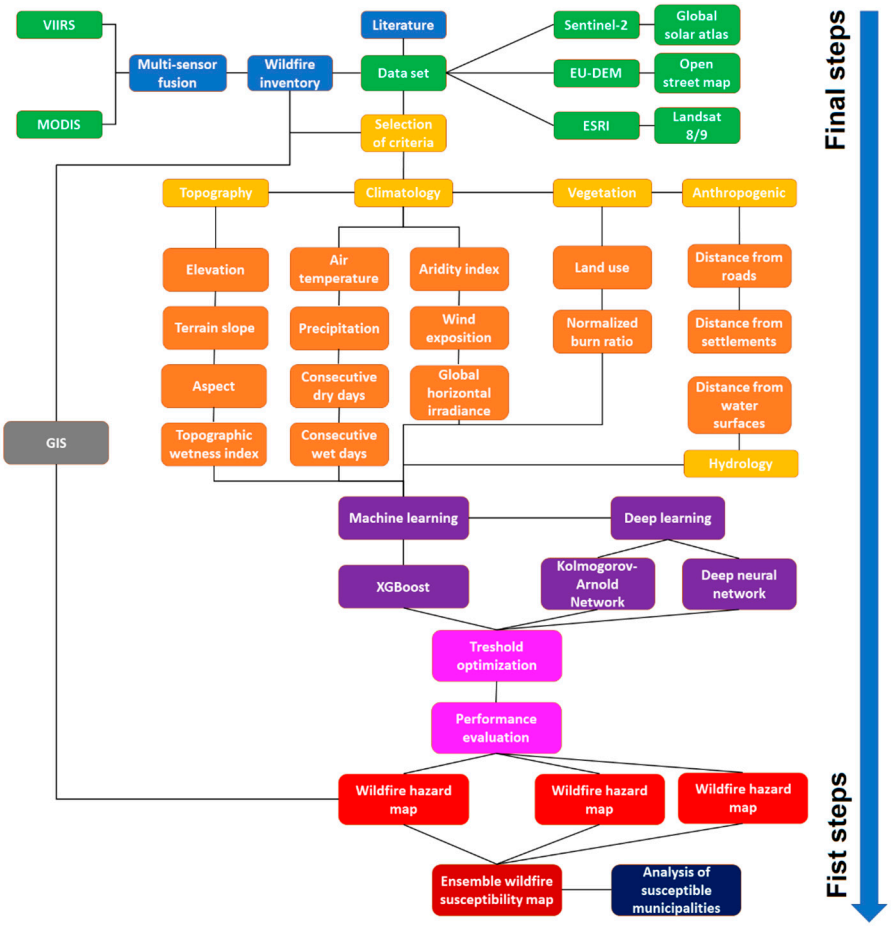


Figure 9. Flowchart of wildfire prediction.

3. Results

3.1. Performance Assessment and Threshold Adjustment

To ensure reliable classification of wildfire-prone areas, each model was subjected to a post-training threshold optimization procedure. Instead of applying the default probability cut-off of 0.5, which may not provide the best trade-off between correctly identifying fire-prone cells and avoiding false alarms, we systematically evaluated thresholds from 0.0 to 1.0 in increments of 0.01. For every candidate threshold, model predictions were binarized, and both accuracy and F1 score were computed. The F1 score was chosen as the primary criterion because it balances precision (avoiding false positives) and recall (capturing true positives), both of which are critical in wildfire prediction.

The resulting curves plot accuracy and F1 score as a function of the decision threshold for each model. In these graphs, accuracy generally decreases more slowly across thresholds, while F1 shows a distinct peak, indicating the threshold where the model best balances sensitivity and specificity. The optimal threshold is marked on each curve with a vertical dashed line and annotation, highlighting the corresponding maximum F1 score.

For the DNN, the threshold curve showed a relatively broad plateau around the optimal point, reflecting stable performance across a range of cut-offs (Figure 10). This indicates that the network is less sensitive to small changes in threshold and provides robust classification behavior.

The KANs produced a sharper F1 peak, meaning their classification performance was more sensitive to the exact choice of threshold (Figure 11). Identifying the best cut-off was particularly important for this model to avoid a drop in precision or recall.

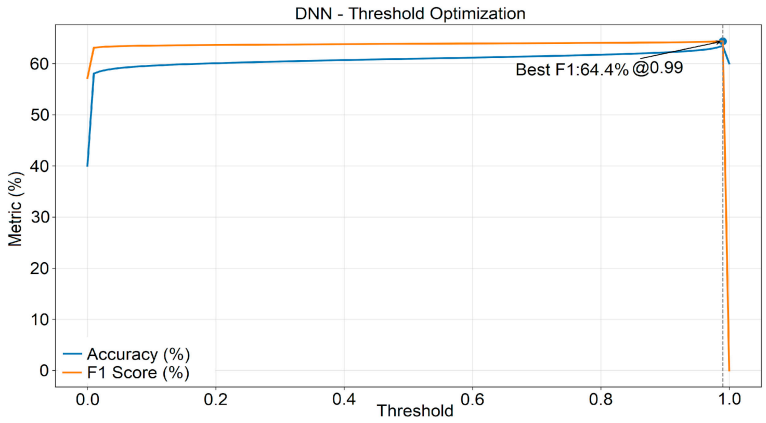


Figure 10. Threshold optimization for the DNN model.

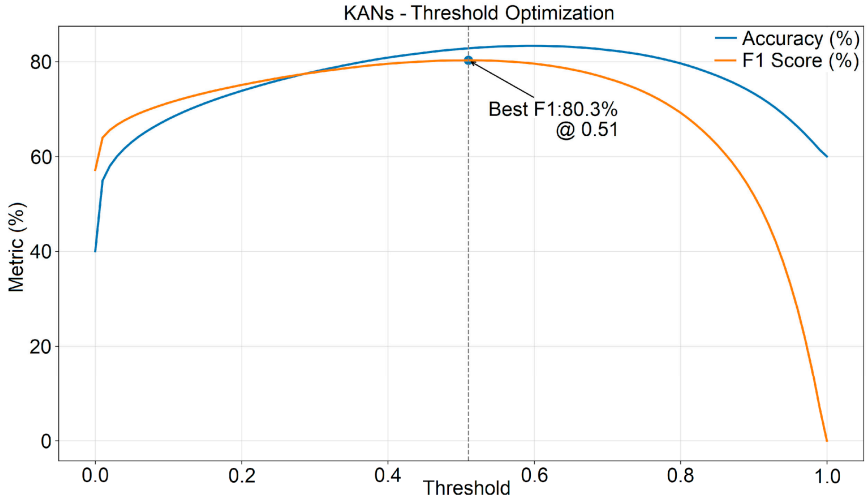


Figure 11. Threshold optimization for the KANs model.

In contrast, the XGBoost model demonstrated a more gradual improvement in F1 with increasing threshold, peaking at a lower value than the deep learning models (Figure 12). This suggests that XGBoost tends to be more conservative in predicting fire-prone areas, and threshold tuning was necessary to improve its balance between false positives and false negatives.

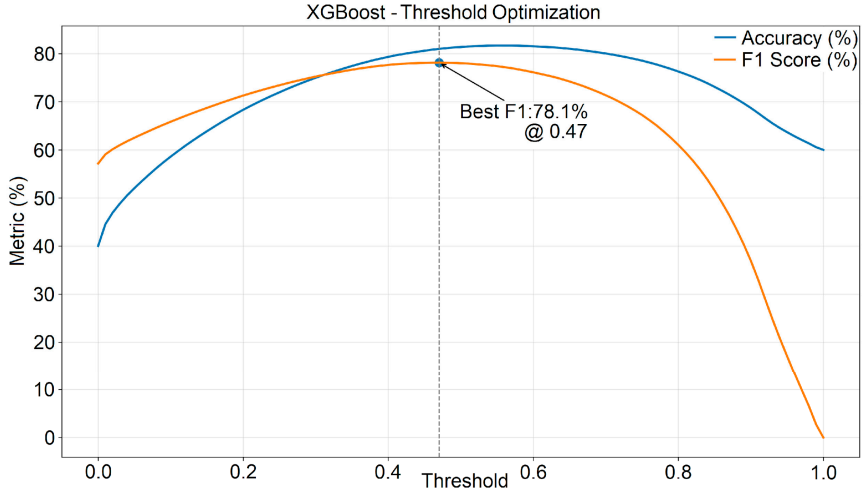


Figure 12. Threshold optimization for the XGBoost model.

Overall, the threshold optimization graphs provided essential guidance for selecting model-specific probability cut-offs that maximize predictive performance. By applying these optimized thresholds, the models achieved more balanced and practically useful wildfire susceptibility classifications than would have been possible with a fixed default threshold.

The matrices show the performance of different models in the spatial prediction of wildfires (Figure 13). With the DNN model, it is noted that it correctly identifies a large proportion of fire-free (48.1%) and fire (35.3%) areas, while having a relatively low number of missed fires (4.7%). However, it also records 11.9% of false alarms, meaning that it often indicates a fire where one does not exist. The KANs model yields very similar results, also with a small number of missed fires, but with a slightly higher share of false alarms (12.7%), which makes it slightly less accurate than DNN in terms of unnecessary alarms.

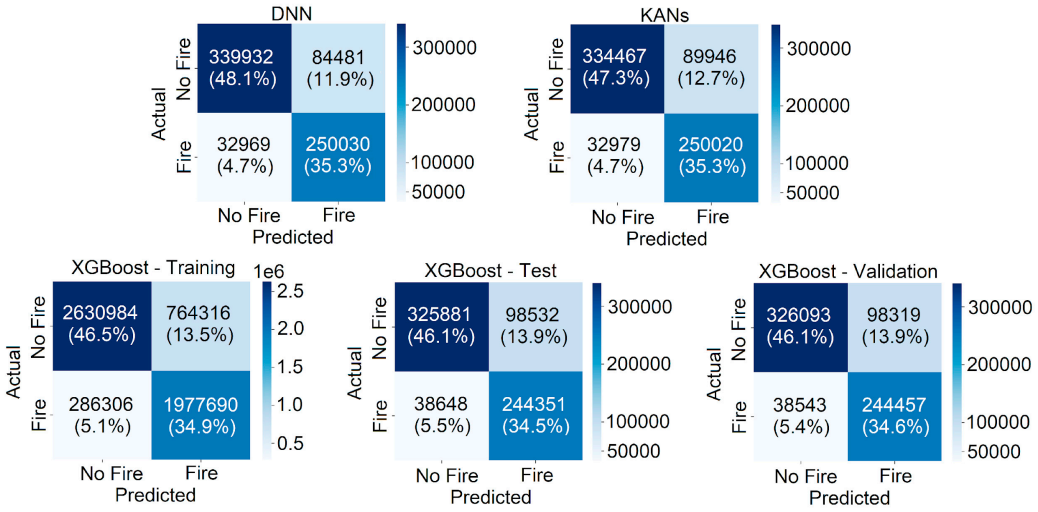


Figure 13. Confusion matrices showing the performance of DNN, KANs, and XGBoost models on training, test, and validation datasets for spatial wildfire prediction.

XGBoost was analyzed on training, test, and validation sets. In training, it shows very good results with a high proportion of correct predictions of fires (34.9%) and areas without fires (46.5%), but with 13.5% of false alarms and 5.1% of missed fires, which indicates the possibility of overlearning. When looking at the test and validation sets, the performance is almost identical and confirms the stability of the model. It correctly identifies approximately 34.5% of fires and 46.1% of fire-free areas, while generating approximately 14% of false alarms and slightly more than 5% of missed cases. It shows that XGBoost performs consistently and generalizes well, but its main drawback is the relatively high number of false positive predictions.

In general, DNN and KANs models have the advantage of missing a very small number of real fires, which is crucial in disaster prevention, although they are prone to a slightly higher number of false alarms. XGBoost, on the other hand, is very stable and reliable across different datasets. However, due to a higher proportion of false alarms, it can generate too many unnecessary warnings in practice. This shows that the choice of model depends on priorities. If the goal is to minimize missed fires, DNN and KANs are better options, while XGBoost is more suitable when stability and robustness are needed on large amounts of data.

3.2. Spatial Analysis and Wildfire Susceptibility Assessment

By processing quantitative variables and an inventory of wildfires using machine learning and a geographic information system, synthetic maps of Serbia's vulnerability to

wildfires were generated. The results clearly indicate that the northern and northeastern parts of Serbia (Vojvodina), as well as certain areas in eastern and southeastern Serbia, are most vulnerable to fires (Figure 14). In Vojvodina, the high risk is associated with large agricultural areas, arid climatic conditions, and flat terrain that facilitate the occurrence and spread of fire. In the southeastern regions, the risk is pronounced due to the combination of vegetation (meadows, and agricultural plots), topography, and climatic conditions. In contrast, the western and southwestern parts of Serbia, particularly regions with a higher share of forests and mountainous terrain, as well as substantial precipitation, indicate a low or very low risk.

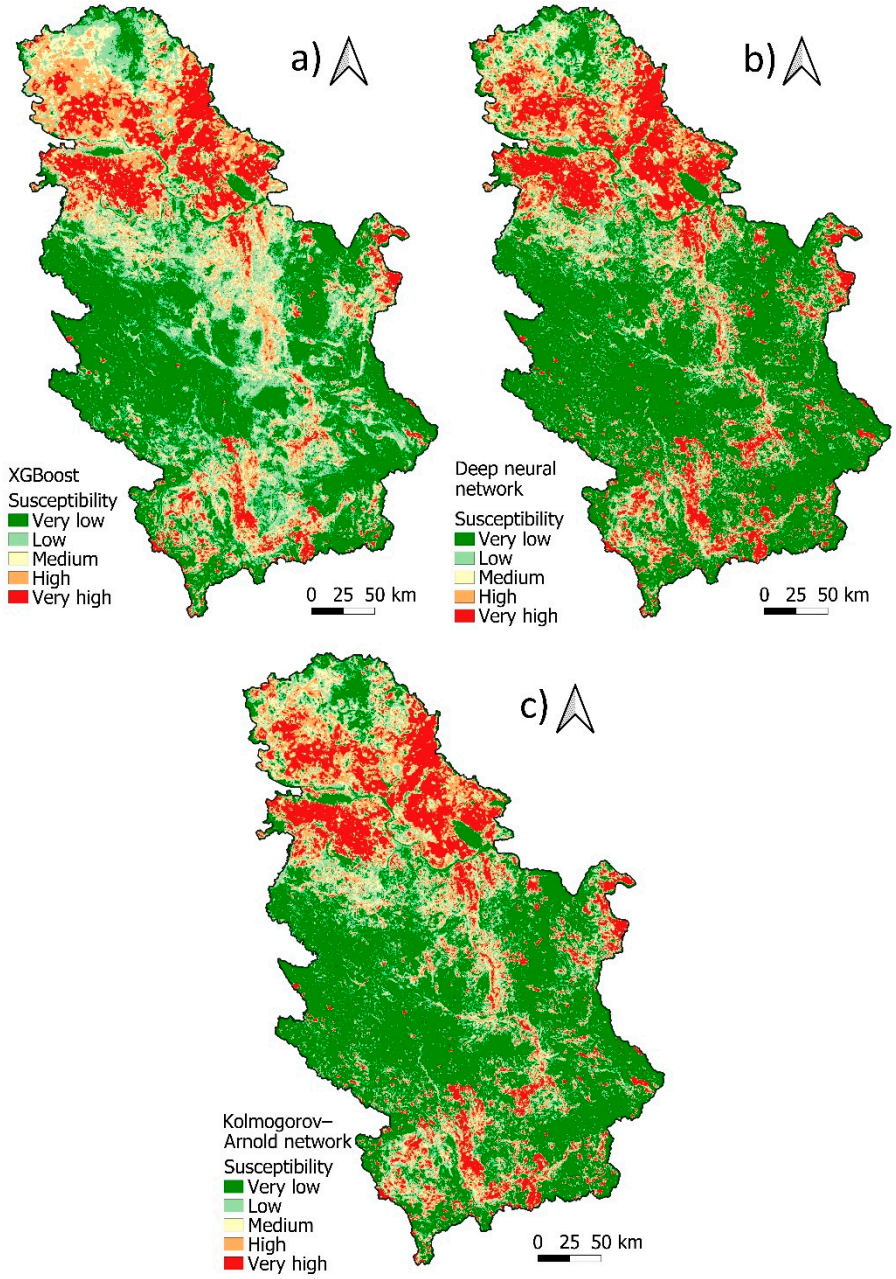


Figure 14. Wildfire hazard maps: (**a**) XGBoost; (**b**) Deep neural network; (**c**) Kolmogorov–Arnold networks.

According to the models used, XGBoost classifies 40% of Serbia's territory as "very low" susceptible, but also shows the largest share of "low" threatened areas (21.3%).

According to this model, 11.5% of Serbia's surface is very highly susceptible to wildfires (Table 2).

Table 2. Spatial distribution of wildfire susceptibility classes in the XGBoost, DNN, KANs, and ensemble models.

Model	Wildfire Susceptibility (%)				
	Very Low	Low	Medium	High	Very High
XGBoost	40.0	21.3	14.7	12.6	11.5
Deep neural network	50.5	11.7	10.9	11.7	15.2
Kolmogorov-Arnold networks	47.3	13.5	12.0	12.3	14.8
Integrated (Ensemble)	46.1	15.7	13.0	12.5	12.7

On the other hand, the Deep neural network marks 50.5% of the territory as "very low" threatened, but at the same time records the highest percentage of areas with "very high" risk (15.2%), which shows that this model tends to emphasize extreme values. The Kolmogorov-Arnold network yields more balanced results, with 47.3% of "very low" vulnerable areas and 14.8% of "very high" vulnerable areas, indicating that it can be considered somewhat more conservative than DNN. The integrated (ensemble) map is generated by averaging the three summary maps. The ensemble model yields values that are intermediate between the individual algorithms: 46.1% of the territory is marked as "very low" and 12.7% as "very high", indicating that the combination of methods moderates the extreme estimates of the individual models (Figure 15).

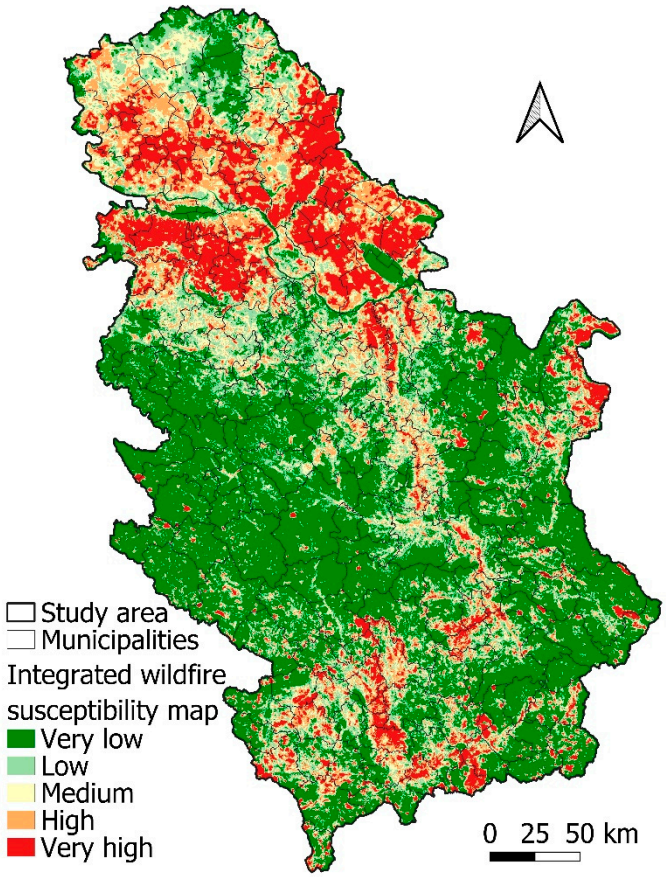


Figure 15. Integrated (ensemble) wildfire susceptibility map.

Overall, the most vulnerable areas of Serbia are the plains and hilly regions of the north and east, while central and western Serbia show a lower risk. The models differ in

the degree to which they rate areas as "very high" threatened, but all confirm the same spatial pattern. This means that, regardless of the method, critical zones can be clearly identified where prevention and protection measures should be directed. In order to analyze Serbia's vulnerability to fires in more detail, results were generated by municipalities based on high and very high susceptibility. Percentage-wise, most of the most vulnerable municipalities are located in Vojvodina, which is not surprising considering the plain character of the terrain, the dominance of agricultural areas, and the dry climatic conditions during the summer.

The most susceptible municipality is Žabalj, where as much as 89.4% of the territory is exposed to a high risk of fire. In second place is Žitište, with 89.3% of the threatened area, followed by Srbobran with 85.8%. Vrbas is also high on the list with 88.2%, while the municipalities of Obilić and Titel record 84.4% each. Nova Crnja (81.8%) and Sečanj (81.7%) are also in the group with a similar level of vulnerability (Table 3). Areas with plain landscapes and large agricultural plots, as well as low forest cover, exhibit the greatest sensitivity to the occurrence and spread of fires. The main reasons for such results lie in a combination of climatic factors (long dry periods, high temperatures, and winds), topography (plain terrain that facilitates the spread of fire), and human activities (proximity to settlements, roads, and intensive agricultural production, which implies the presence of easily flammable vegetation).

Table 3. Spatial distribution of wildfire susceptibility (very high) in Serbia by municipality. Note: M—Municipality; TA—Total area; SA—Susceptible areas; %—Percentage of susceptibility of the municipality.

M	TA (km^2)	SA (km^2)	%	M	TA (km^2)	SA (km^2)	%
Bosilegrad	571.2	89.6	15.7	Kovin	736.1	392.2	53.3
Kanjiža	398.5	4.4	1.1	Pančevo	755.8	565.0	74.8
Kladovo	627.2	250.1	39.9	Prijepolje	827.4	29.4	3.6
Titel	260.8	220.0	84.4	Sjenica	1058.5	83.6	7.9
Bečej	486.5	94.6	19.4	Vračar	2.9	0	0
Žabalj	399.8	357.6	89.4	Kikinda	782.6	300.7	38.4
Alibunar	601.0	470.1	78.2	Čoka	321.3	56.6	17.6
Kovačica	418.6	330.7	79.0	Ada	227.0	0.3	0.1
Temerin	169.6	114.7	67.6	Novi Bečej	608.6	210.2	34.5
Kula	481.4	367.6	76.4	Zrenjanin	1326.1	961.4	72.5
Novi Kneževac	305.4	51.0	16.7	Žitište	525.0	468.8	89.3
Novi Sad	699.2	431.4	61.7	Nova Crnja	272.6	223.1	81.8
Mali Iđoš	181.2	74.6	41.2	Kuršumlija	951.7	101.2	10.6
Senta	293.4	2.7	0.9	Sevojno	19.6	1.3	6.9
Bačka Topola	596.0	96.3	16.2	Požarevac	378.6	214.6	56.7
Sremski Karlovci	50.6	2.8	5.6	Lebane	336.8	62.4	18.5
Inđija	384.6	235.2	61.2	Medveđa	524.3	67.5	12.9
Bogatić	384.3	242.5	63.1	Subotica	1007.4	64.2	6.4
Ljubovija	356.2	4.1	1.1	Smederevska Palanka	421.3	109.5	26.0
Vrbas	375.5	331.4	88.2	Petrovac na Mlavi	655.1	17.3	2.6
Bački Petrovac	158.3	117.8	74.4	Brus	605.9	11.5	1.9
Beočin	184.2	67.9	36.9	Raška	670.2	9.2	1.4
Ruma	582.0	461.9	79.4	Aleksandrovac	386.6	8.0	2.1
Rakovica	30.0	4.9	16.2	Novi Pazar	742.5	51.5	6.9
Stara Pazova	344.5	251.9	73.1	Tutin	741.7	54.1	7.3
Trgovište	370.6	32.9	8.9	Srbobran	284.1	243.6	85.8
Opovo	203.3	162.4	79.9	Vlasotince	307.9	14.6	4.7
Bujanovac	460.9	200.5	43.5	Gadžin Han	324.7	35.6	10.9
Surdulica	628.4	138.7	22.1	Kostolac	101.4	42.0	41.4
Malo Crniće	269.5	97.2	36.1	Barajevo	212.9	9.4	4.4

Table 3. *Cont.*

M	TA (km^2)	SA (km^2)	%	M	TA (km^2)	SA (km^2)	%
Kučevo	721.2	135.9	18.8	Lazarevac	383.0	37.5	9.8
Žagubica	760.1	37.4	4.9	Obrenovac	409.7	35.1	8.6
Bor	856.3	160.0	18.7	Čukarica	157.0	53.4	34.0
Ćuprija	288.0	106.7	37.1	Vladimirci	337.6	62.4	18.5
Despotovac	623.2	114.5	18.4	Pirot	1232.1	90.3	7.3
Svilajnac	326.1	79.1	24.2	Lajkovac	185.2	25.0	13.5
Koceljeva	257.5	27.2	10.6	Ub	456.2	54.1	11.9
Blace	306.2	10.2	3.3	Surčin	288.6	151.9	52.6
Preševo	264.7	126.9	47.9	Pećinci	488.7	367.7	75.2
Osečina	318.6	2.8	0.9	Sombor	1216.4	482.0	39.6
Valjevo	905.1	12.3	1.4	Apatin	380.5	160.8	42.3
Krupanj	341.7	3.3	1.0	Odžaci	411.0	295.2	71.8
Jagodina	469.5	104.5	22.3	Bač	367.4	142.2	38.7
Batočina	135.7	13.9	10.3	Bačka Palanka	589.7	389.0	66.0
Rača	215.6	14.9	6.9	Nova Varoš	581.4	18.5	3.2
Kragujevac	834.7	87.4	10.5	Priboj	552.9	33.0	6.0
Knić	413.2	54.9	13.3	Bela Palanka	516.9	60.0	11.6
Rekovac	366.0	9.7	2.7	Knjaževac	1202.2	46.2	3.8
Topola	356.6	16.4	4.6	Svrljig	497.2	64.4	13.0
Gornji Milanovac	836.4	11.9	1.4	Ražanj	288.7	9.8	3.4
Stari Grad	5.4	0	0	Paraćin	541.3	91.1	16.8
Savski Venac	14.1	0	0	Boljevac	827.7	46.0	5.6
Novi Beograd	40.7	10.0	24.6	Sokobanja	525.4	26.0	4.9
Zemun	149.8	92.2	61.6	Aleksinac	706.9	134.1	19.0
Kosjerić	358.6	1.8	0.5	Kruševac	854.0	59.8	7.0
Arilje	349.1	3.2	0.9	Ćićevac	123.6	25.1	20.3
Ivanjica	1089.8	17.0	1.6	Prokuplje	758.9	127.6	16.8
Trstenik	448.1	12.5	2.8	Žitorađa	213.9	96.0	44.9
Varvarin	249.4	76.2	30.5	Merošina	193.1	25.5	13.2
Vrnjačka Banja	238.6	4.9	2.0	Doljevac	121.2	53.9	44.5
Sečanj	522.6	426.7	81.7	Leskovac	1025.0	122.8	12.0
Požega	426.1	9.6	2.3	Bojnik	263.9	113.5	43.0
Čačak	636.4	14.1	2.2	Palilula (Niš)	116.6	24.3	20.8
Voždovac	148.4	7.7	5.2	Medijana	10.8	0.9	8.1
Grocka	299.7	39.7	13.3	Niška Banja	146.2	3.0	2.0
Zvezdara	31.1	3.6	11.5	Pantelej	141.0	10.3	7.3
Mladenovac	339.0	34.4	10.2	Crveni Krst	181.8	32.3	17.8
Palilula (Beograd)	450.7	114.7	25.5	Srbica	375.2	105.3	28.1
Sopot	270.7	14.9	5.5	Kosovo Polje	99.5	69.8	70.2
Smederevo	484.3	290.6	60.0	Obilić	106.9	90.2	84.4
Babušnica	528.6	20.7	3.9	Glogovac	289.6	32.9	11.4
Golubac	367.3	20.2	5.5	Klina	401.8	195.8	48.7
Majdanpek	931.8	39.7	4.3	Lipljan	406.0	211.0	52.0
Negotin	1089.8	451.6	41.4	Novo Brdo	80.8	5.2	6.5
Zaječar	1069.5	157.1	14.7	Kosovska Kamenica	520.5	133.5	25.7
Dimitrovgrad	483.1	59.5	12.3	Priština	564.3	92.6	16.4
Crna Trava	312.0	1.9	0.6	Uroševac	350.2	98.2	28.0
Šid	686.9	369.8	53.8	Štimlje	136.5	26.6	19.5
Mali Zvornik	183.9	1.5	0.8	Suva Reka	431.0	101.4	23.5
Bajina Bašta	673.3	12.3	1.8	Orahovac	400.6	78.6	19.6
Plandište	383.2	289.0	75.4	Peć	605.2	111.5	18.4
Vršac	799.1	555.2	69.5	Istok	455.8	119.8	26.3
Sremska Mitrovica	761.2	581.0	76.3	Dečani	371.0	46.2	12.5
Šabac	797.3	243.7	30.6	Đakovica	587.6	217.8	37.1
Loznica	612.0	29.6	4.8	Prizren	760.0	134.2	17.7
Bela Crkva	353.4	137.7	39.0	Gora	309.8	77.9	25.1
Veliko Gradište	342.9	103.6	30.2	Štrpce	232.7	36.9	15.9
Kraljevo	1529.9	30.2	2.0	Kačanik	304.8	42.1	13.8
Užice	647.5	17.8	2.7	Vitina	290.9	145.1	49.9
Ljig	278.7	6.8	2.4	Gnjilane	517.8	151.3	29.2

Table 3. *Cont.*

M	TA (km^2)	SA (km^2)	%	M	TA (km^2)	SA (km^2)	%
Aranđelovac	375.8	13.0	3.5	Zubin Potok	334.3	54.3	16.2
Mionica	329.6	5.9	1.8	Vučitrn	346.8	195.2	56.3
Lapovo	54.9	7.4	13.4	Kosovska Mitrovica	346.0	98.2	28.4
Irig	230.1	138.2	60.1	Podujevo	622.9	275.4	44.2
Žabari	263.8	90.1	34.2	Zvečan	116.5	18.1	15.6
Vladičin Han	365.8	25.1	6.8	Leposavić	538.6	90.2	16.8
Velika Plana	345.2	124.1	36.0	Vranje	600.0	99.1	16.5
Lučani	454.7	2.9	0.6	Vranjska Banja	258.4	15.3	5.9
Čajetina	646.6	17.3	2.7				

On the other hand, the least threatened municipalities in Serbia are the city municipalities of Belgrade, specifically Savski Venac and Vračar, both of which have 0% of their areas at risk (Figure 16).

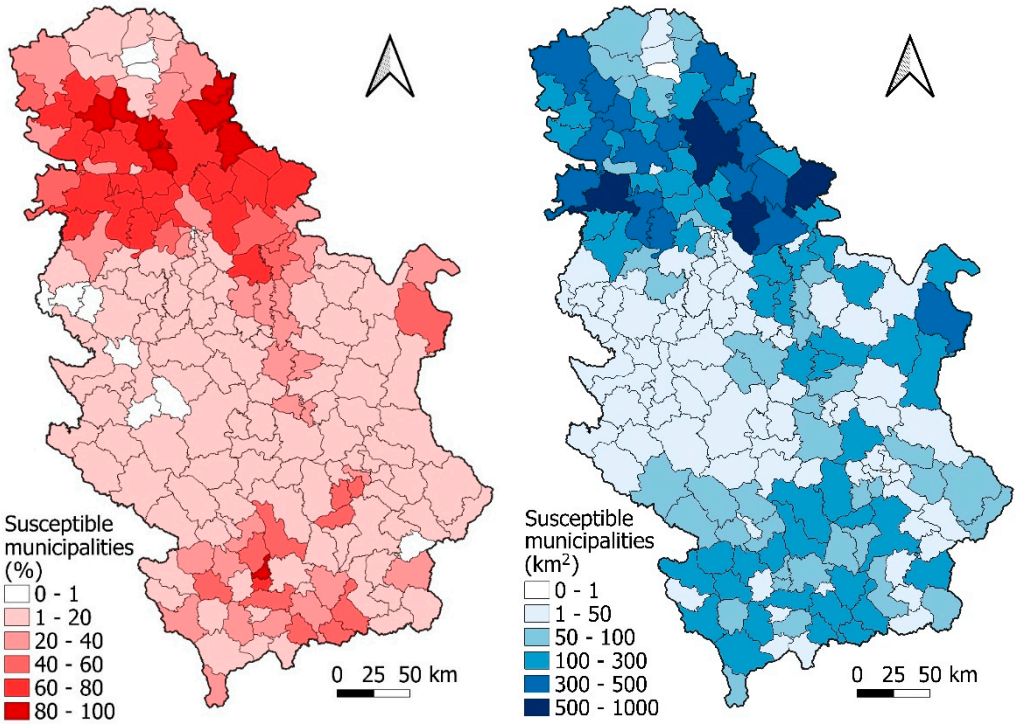


Figure 16. Very high susceptibility of municipalities in percentage (**left**) and square kilometers (**right**).

This is expected because these are highly urbanized territories, with almost no agricultural or forest areas that would be susceptible to wildfire spread. The situation is similar to Stari Grad, where the share of endangered areas is negligible. When analyzing the less urbanized municipalities located in the lower part of the table, it is noted that the percentage of risk and the absolute area at risk are also very low. For example, in the municipality of Ada, out of a total of 227 km^2, only 0.3 km^2 falls into the high-risk category, which is only 0.1% of the territory. Lučani and Crna Trava each have 0.6%, and Kosjerić has 0.5%. Although these are hilly areas where fires can occur, their spatial distribution and the combination of natural factors (climate, vegetation, topography) result in a lower threat compared to the lowland municipalities of Vojvodina.

Low values do not mean that the danger of fire does not exist, but rather that the statistical risk is significantly lower and that the potentially threatened areas are confined to limited parts of the municipality. In practice, this means that the local governments of these municipalities do not have to invest significant funds for the prevention and

control of wildfires in areas with a high percentage of threats. However, they should still maintain basic capacities and preventive measures because even small fires can have serious consequences on the local ecosystem and environment.

3.3. Evaluation of Model Accuracy and Predictive Power

The comparative evaluation of the three applied models provides insights into their predictive capabilities in assessing wildfire susceptibility across Serbia. To ensure a robust evaluation and account for spatial autocorrelation, model performance was assessed using spatial cross-validation (CV) with five disjoint folds. For each fold, ROC and precision–recall (PR) curves were generated to visualize the consistency of predictive performance across spatial partitions. Ninety-five percent confidence intervals for the main evaluation metrics (accuracy, F1-score, PR-AUC, and ROC-AUC) were computed using bootstrap resampling (n = 1000) over folds. Furthermore, predictive uncertainty maps were produced by calculating the variance of susceptibility probabilities across folds, highlighting areas of higher model disagreement and uncertainty.

Several performance metrics were used, including test accuracy, F1-score for the positive class, Precision–Recall Area Under the Curve (PR-AUC), Receiver Operating Characteristic Area Under the Curve (ROC-AUC), and the optimal classification threshold (Table 4).

Table 4. Comparison of predictive performance between DNN, KANs, and XGBoost models.

Model	Test Accuracy	F1-Score (Positive Class)	PR-AUC	ROC-AUC
DNN	0.834	0.8098	0.8772	0.9228
KANs	0.8061	0.781	0.8533	0.9008
XGBoost	0.8262	0.8027	0.8708	0.9178

The DNN model achieved the strongest overall performance, with a test accuracy of 0.834, the highest F1-score (0.8098), PR-AUC (0.8772), and ROC-AUC (0.9228). These results indicate that the deep learning approach was most effective in balancing sensitivity and specificity while maintaining high discriminative power. The optimal threshold of 0.50 indicates that a standard probability cutoff yielded the best results, reflecting the robustness of the DNN in distinguishing between fire-prone and non-fire-prone areas. The XGBoost model, although slightly underperforming compared to DNN, still demonstrated competitive results, with a test accuracy of 0.8262, an F1-score of 0.8027, a PR-AUC of 0.8708, and an ROC-AUC of 0.9178. These values suggest that the tree-based ensemble method was particularly effective at capturing complex nonlinear relationships within the data. Notably, XGBoost performed well in terms of precision–recall dynamics, highlighting its strength in handling class imbalance, which is typical in wildfire susceptibility datasets. The KANs model performed slightly weaker than DNN and XGBoost, with a test accuracy of 0.8061, an F1-score of 0.781, a PR-AUC of 0.8533, and an ROC-AUC of 0.9008. While still effective, the lower F1-score suggests that KANs had more difficulty in balancing false positives and false negatives. Interestingly, the optimal classification threshold for KANs was 0.45, indicating that this model required a more conservative cutoff to achieve its best trade-off between sensitivity and precision, possibly due to its higher sensitivity to long-term climatic sequences, as highlighted in the SHAP analysis.

Overall, the results show that DNN outperformed the other models, providing the most reliable classification of wildfire susceptibility zones in Serbia. XGBoost ranked second, offering competitive performance with strong precision–recall balance, while KANs, although effective, demonstrated slightly lower predictive stability. The consistency across PR-AUC and ROC-AUC values (>0.85 and >0.90 for all models, respectively) confirms

the robustness of the modeling approach. It highlights the potential of machine learning methods for operational wildfire risk mapping.

3.4. Interpretation of Feature Importance

To better understand the internal decision-making of the applied models, we employed SHAP (SHapley Additive exPlanations) analysis. SHAP values quantify the contribution of each predictor variable to the final classification, allowing both the ranking of influential factors and insight into the direction of their impact. Figures 17–19 summarize the SHAP results for the DNN, KANs, and XGBoost models. Importantly, several predictors exhibit non-monotonic and interaction-dependent effects. In the SHAP summaries (Figures 17–19) low feature values cluster around ≈ 0 SHAP (weak influence), whereas high values spread to both negative and positive SHAP. This pattern indicates that the same high value can either increase or decrease susceptibility depending on accompanying conditions (e.g., land use, aridity, fuel continuity), which the models capture through nonlinearities and interactions.

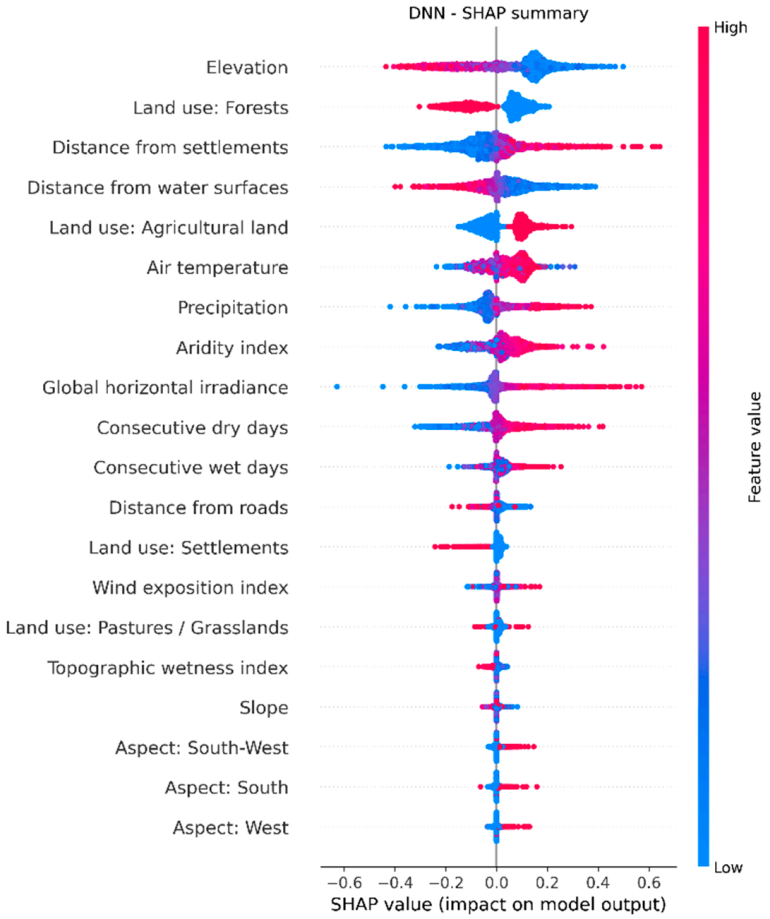


Figure 17. SHAP analyses for the DNN model for wildfire risk assessment in Serbia.

For the DNN, the most influential predictors were elevation, land use (forests and agricultural land), and distance from settlements and water surfaces. Climatic variables such as air temperature and precipitation also played a central role. Lower elevation values increase fire susceptibility, while higher elevations tend to reduce it. Similarly, areas closer to water surfaces show higher susceptibility, likely due to human activity in surrounding agricultural and settlement zones, whereas a greater distance from water reduces susceptibility. Agricultural land and settlements increased fire risk, consistent with anthropogenic ignition factors (Figure 17).

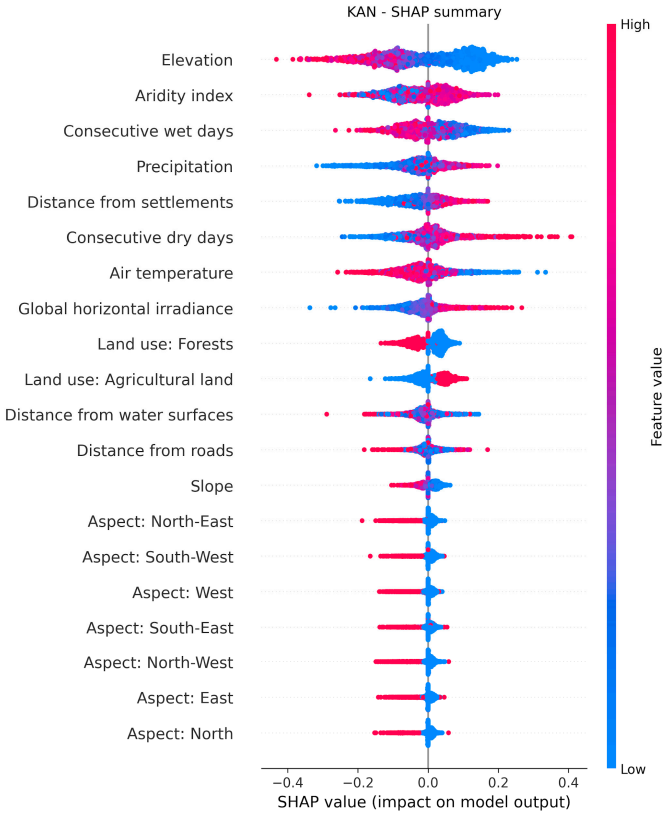


Figure 18. SHAP analyses for the KANs model for wildfire risk assessment in Serbia.

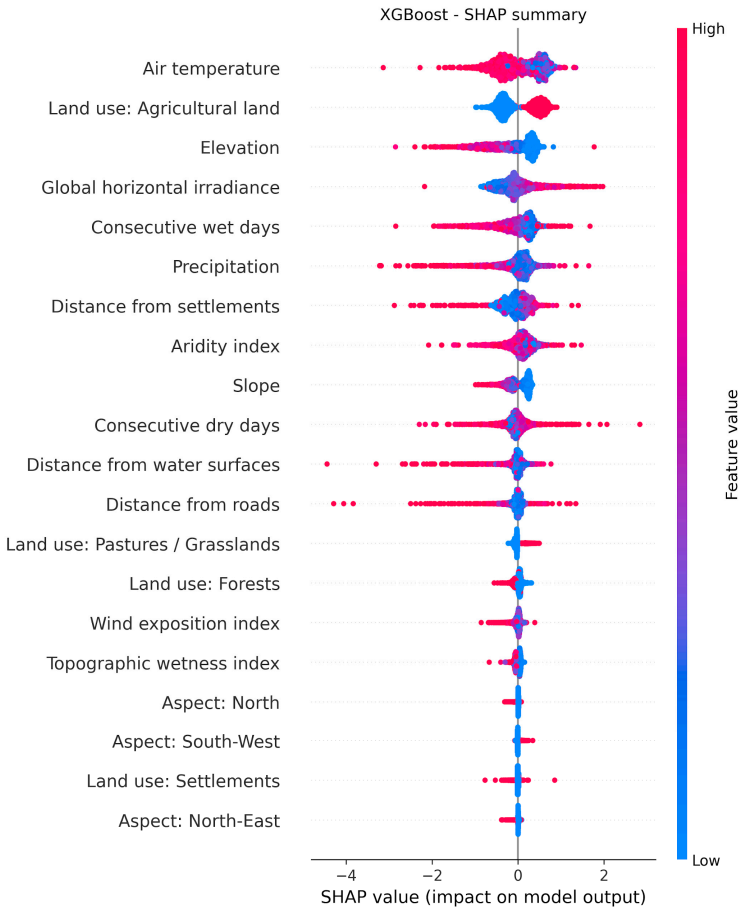


Figure 19. SHAP analyses for the XGBoost model for wildfire risk assessment in Serbia.

The KANs model produced a slightly different ranking, with elevation, aridity index, consecutive wet days, and precipitation among the most critical predictors. Compared to the DNN, KANs gave higher importance to long-term climatic indicators such as dryness and wetness sequences (Figure 18). This reflects the model's sensitivity to sustained climatic conditions rather than isolated factors. Anthropogenic variables (distance from settlements, distance from roads) were also significant, reinforcing the influence of human presence on wildfire occurrence.

In the case of XGBoost, the dominant variables were air temperature, agricultural land, elevation, and global horizontal irradiance. This indicates that the tree-based model placed greater weight on direct climatic and land use factors. XGBoost also highlighted consecutive wet days and precipitation as important, but generally ranked anthropogenic features lower than the deep learning models (Figure 19). Its variable importance profile suggests a focus on short-term environmental stressors that directly affect vegetation flammability.

Across all three models, several consistent factors emerged as the primary drivers of wildfire susceptibility in Serbia: topography (elevation, slope, aspect), climatic conditions (temperature, precipitation, dry and wet days, aridity index, irradiance, wind exposure), land use (forests, agricultural land, pastures), and anthropogenic influences (distance from roads and settlements). The agreement between models on these key variables underscores their fundamental role in shaping wildfire dynamics.

Some predictors show non-monotonic, context-dependent effects. In the SHAP plots (Figures 17–19), low values have little impact (clustered near zero), while high values can either increase or decrease susceptibility depending on other conditions (e.g., land use, aridity, fuel availability). This reflects nonlinearities and interactions in the models.

For consecutive dry days (CDD), higher values often increase fire risk in fuel-rich areas, as drying promotes ignition. In very arid or sparsely vegetated zones, however, extremely high CDD may reduce predicted risk because of limited available fuel. This explains the mixed SHAP values at the upper range. Distance from roads shows a U-shaped effect. Close to roads, fires may be suppressed quickly or blocked, while at intermediate distances human ignitions are more frequent, raising risk. At larger distances ignition pressure decreases again, unless combined with continuous, flammable vegetation. This explains the bidirectional SHAP patterns.

However, the relative importance and interpretation varied, with deep learning models emphasizing a broader mix of anthropogenic and climatic drivers, while XGBoost concentrated more narrowly on climate and land cover.

To evaluate the stability and consistency of model interpretability, a cross-model concordance analysis was performed using Spearman's rank correlation (ρ) between the SHAP-derived feature importance scores of the DNN, KAN, and XGBoost models. The results (Table 5) show high to very high agreement among all model pairs ($\rho = 0.817–0.886$), indicating that the three architectures identify largely the same set of dominant predictors. The highest concordance ($\rho = 0.886$) was observed between DNN and XGBoost, suggesting that despite their different learning mechanisms, both capture similar spatial and climatic patterns influencing the target variable. Slightly lower correlations between KAN and the other two models ($\rho \approx 0.82–0.83$) can be attributed to architectural biases—KAN tends to model smooth continuous dependencies, whereas XGBoost captures sharp, threshold-based relationships. Nevertheless, the consistently high ρ values confirm that the SHAP-based feature importance rankings are robust across model families, reinforcing the reliability of the identified key environmental drivers.

Table 5. Cross-model concordance of feature importance rankings.

Model Pair	Spearman's ρ (Feature Importance Concordance)	Interpretation
DNN—KAN	0.817	Strong agreement—both neural models emphasize similar climatic and anthropogenic drivers.
DNN—XGBoost	0.886	Very strong agreement—consistent feature prioritization across neural and tree-based architectures.
KAN—XGBoost	0.834	Strong agreement—minor divergences reflecting architectural biases in handling nonlinear interactions.

Figure 20 illustrates the key input variables utilized in the DNN and KANs models for predicting wildfire occurrences and their corresponding relative importance. Both models indicate that climate variables play the most significant role. Global horizontal irradiance, which measures the amount of solar energy on the surface, is shown to be the most influential factor because it directly affects the drying of vegetation and increases the risk of flammability. In addition, air temperature and the aridity index, which combines precipitation and evapotranspiration, further explain how dry and fire-prone an area is. Precipitation and the number of consecutive dry or wet days also play a significant role, as continuous dry periods create conditions favorable for the spread of fire, while wet periods reduce the risk.

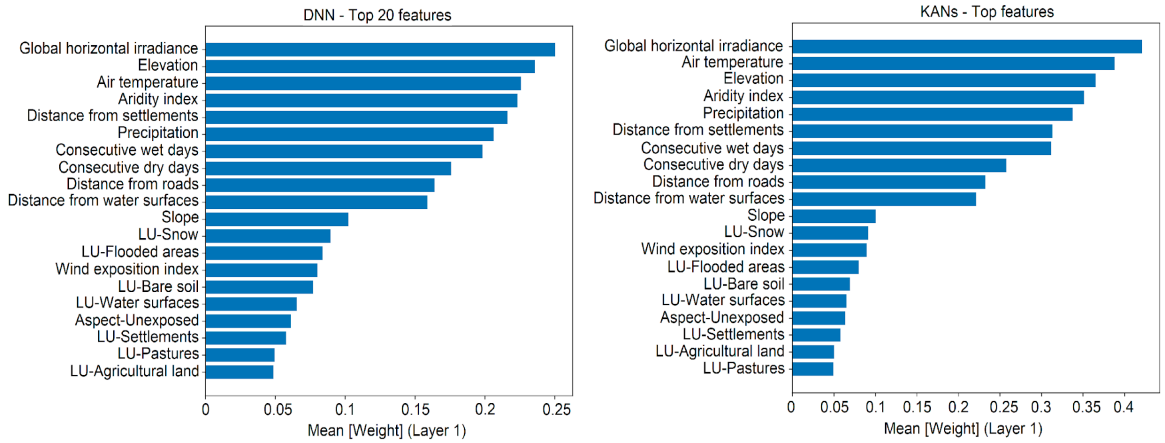


Figure 20. Top contributing features for wildfire prediction using the DNN and KANs models.

Topographical factors are another important group of variables that influence the prediction. Altitude determines the type of vegetation and climatic conditions, and the terrain slope affects the speed of flame spread. Meanwhile, the exposure of the terrain and the wind index alter the microclimatic conditions and the availability of fuel. Snow and flooded areas reduce the risk by acting as barriers to the spread of fire.

The third group consists of anthropogenic and hydrographic variables. Distance from settlements and roads is important because human activities are one of the main causes of fires. Distance from water surfaces reduces the likelihood of fire because areas closer to water have higher humidity. Land use, such as agricultural land, pastures, and forests, also affects the risk because different types of land have varying flammability and resistance to fire spread.

Both models recognize the same groups of factors as important, but rank them differently. The DNN model distributes importance more evenly between climatic, topographic, and anthropogenic factors, while KANs emphasizes climatic variables more strongly, especially solar radiation and temperature. It shows that climatic conditions are the primary

driver of the spatial distribution of fires, while relief and human influence contribute to additional explanation and local variations in risk.

Figure 21 presents the top 20 features used by the XGBoost model for wildfire prediction, measured by average gain. The results clearly show that agricultural land is by far the most influential factor, with a much higher contribution than any other variable. Pastures and slope follow as relevant predictors, while forests and water surfaces also play a role, reflecting the influence of land cover on fire occurrence and spread.

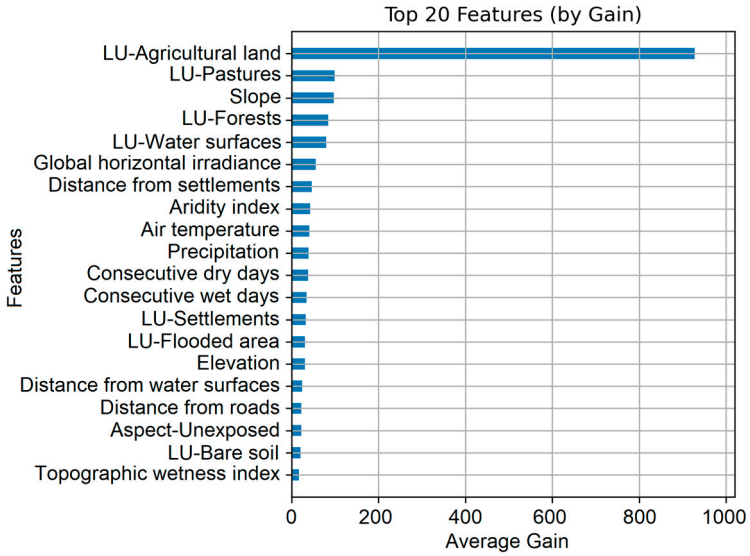


Figure 21. Top contributing features for wildfire prediction using the XGBoost model.

Climatic variables, such as irradiance, temperature, precipitation, and aridity index, contribute significantly less to this model, as do human-related factors, including distance from settlements and roads. This suggests that XGBoost relies heavily on land use information, particularly agricultural areas, with topography and natural cover types having secondary importance, while climate and anthropogenic influences are less emphasized. The XGBoost model ranked agricultural land as by far the most important predictor, with topography and other land cover types playing secondary roles, and climatic and anthropogenic factors contributing less. In contrast, the DNN and KANs models distributed importance more evenly, giving greater weight to climatic variables such as temperature, precipitation, and aridity. This divergence likely reflects the different ways the models capture feature interactions: tree-based ensembles such as XGBoost prioritize strong categorical splits (e.g., land use), while neural networks are more sensitive to continuous variables and complex interactions. Similar contrasts between tree-based and deep learning approaches have been observed in other wildfire studies, where XGBoost and related models emphasized vegetation or land use [18], whereas deep learning methods highlighted climatic and fuel-related factors [9,22]. These differences indicate that combining both model families provides complementary insights into wildfire drivers.

The results and significance of predictors vary across studies depending on the application of machine and deep learning models.

Zhang et al. (2023) investigated the factors influencing the spread of forest fire-affected areas in Liangshan Prefecture, Sichuan, China [64]. An ensemble approach was applied using the GWO-XGBoost model. According to the SHAP analysis, the most significant factors affecting the expansion of burned areas are the Ignition Component (IC), Monthly Temperature (MT), and Population Density (PD). IC, derived from the U.S. National Fire Danger Rating System (NFDRS), proved to be the most important predictor, while monthly temperature and population density also had a strong impact. Other factors, such as the

Fine Fuel Moisture Code (FFMC), Drought Code (DC), altitude, and NDVI, showed smaller but still meaningful contributions. The study concludes that meteorological conditions (particularly temperature), fuel flammability, and human activity (population density) are the main drivers of fire spread, whereas topographic factors play a secondary role.

He et al. (2024) employed a ConvLSTM model to capture spatiotemporal patterns in the data more effectively [24]. Their results show that the model relies most heavily on meteorological variables, with precipitation having by far the greatest impact across all performance indicators. Solar radiation and evapotranspiration also play a role, though to a lesser extent. Topographic and anthropogenic factors such as elevation, slope, distance to roads or settlements, and population contribute very little, with their effect being almost negligible. Vegetation indices make a limited contribution, with the green NDVI channel showing a slightly higher influence, although it remains relatively small. Overall, the findings indicate that climatic variables dominate in explaining and predicting the studied phenomenon, while physical and human spatial characteristics have comparatively minor importance.

Jiang et al. (2024) used a deep learning-based CNN model to predict the probability of fire occurrence [9]. The most influential variables on the occurrence of fires were relative humidity, temperature, NDVI (a vegetation index), and precipitation. At the same time, socioeconomic factors, such as GDP and population density, had a much smaller impact. Relative humidity stood out as the most significant contributing factor, indicating that fires in the region are strongly associated with dry conditions and low atmospheric humidity. Temperature and NDVI are additional contributors, as high temperatures and lush vegetation increase the risk of flammability, while precipitation reduces the likelihood of fire outbreaks. On the other hand, topographical variables such as altitude, slope, and aspect, as well as climatic oscillations such as daily temperature range and wind speed, had a marginal impact. The authors emphasize that the contribution of these factors varies depending on seasonal conditions. However, on average, meteorological factors (especially humidity and temperature) significantly surpass topographical and socioeconomic factors in explaining fire dynamics.

Bahadori et al. (2023) employed fire susceptibility mapping models utilizing a deep learning approach with various recurrent neural networks (RNN, LSTM, and ConvLSTM) [22]. In the study, they combined data from two different satellite sources, MODIS and Landsat-8, to estimate the factors influencing wildfires in 2021. The most significant factors contributing to the occurrence of wildfires include precipitation, temperature, and wind speed. These parameters exhibit a high level of variability and interdependence, confirming that meteorological conditions are of crucial importance in explaining fire dynamics. Other factors, such as altitude, slope, distance from roads and settlements, as well as the vegetation index NDVI, have lower VIF values, indicating a minor but still significant contribution to the models. Generally, the primary risk factors are climatic variables, while topographical and human components are of secondary importance.

Based on the analyzed studies, it is clearly demonstrated that meteorological variables (precipitation, temperature, humidity, and wind speed) are the primary drivers of wildfire occurrence and susceptibility. In contrast, topographic, vegetation, and socioeconomic factors play a much smaller role, and their influence is often secondary. These findings underscore the crucial importance of incorporating climate conditions into predictive models for accurate risk assessment and effective fire management.

4. Discussion

Deep Learning in Wildfire Prediction: Achievements and Limitations

Over the past five years, a large number of deep learning models have been developed and deployed, demonstrating significantly better performance for wildfire prediction than baseline and ensemble machine learning models.

Jin et al. (2020) used convolutional neural networks (CNNs) and a graph convolutional network (GCN) to extract spatial features [65]. The obtained data were forwarded to the GRU recurrent neural network and then reconstructed using deconvolution layers, which achieved semantic segmentation and prediction of fires in urban areas. Compared to classical machine learning methods, such as Random Forest and XGBoost, this deep model has shown significant improvements. Li et al. (2021) took a step further towards spatio-temporal modeling [66]. They developed a CNN-LSTM architecture with an added attention mechanism. CNN served for spatial processing, LSTM for temporal sequences, while attention allowed certain temporal states to be given different importance.

The results showed that this deep learning approach outperforms classical machine learning methods (SVM, XGBoost) and simpler neural networks (basic RNN and LSTM). Zhang et al. (2022) compared different deep CNN-LSTM architectures and concluded that a combination with two convolutional and two LSTM layers represents the optimal balance between accuracy and efficiency in predicting global monthly maps of burned areas [67]. On the other hand, Yoon and Voulgaris (2022) developed a model where CNN reduces the dimensionality of input data, GRU processes temporal sequences, and deconvolution layers restore spatial resolution, enabling risk prediction through time horizons up to four weeks in advance [68]. Both studies clearly belong to the domain of deep learning. Huot et al. (2022) also used deep learning, specifically a convolutional autoencoder and residual UNet [69]. Their goal was spatial prediction—both the probability of fires for the next day and the final burnt area. The results showed that the autoencoder outperforms UNet, but both models outperform standard machine learning approaches.

During 2023, further development of complex deep architectures was observed. Marjani et al. developed a spatio-temporal model that uses a CNN to process different types of data (hourly, daily and constant inputs), while an RNN combines the results and generates risk maps [70]. Their model proved to be more accurate than earlier CNN methods. Bhowmik et al. (2023) presented an innovative U-shaped model (UNet + LSTM). The CNN part (UNet encoder) processes spatial information, LSTM temporal sequences, and the UNet decoder produces detailed risk maps [71]. In California, this deep model achieved over 97% accuracy in predicting large fires, significantly outperforming CNN as a stand-alone method.

Recent work from 2024 has continued to advance spatio-temporal approaches. Rösch et al. applied a graph convolutional network (GCN) in combination with GRU and developed a spatio-temporal graph neural network (ST-GNN) [72]. Their models used complex data on weather, fuel type, and terrain, but had a problem with a high rate of false positive predictions. Although they are complex, these models also belong to deep learning because they rely on neural networks and graph structures, not on classical methods. Jiang et al. (2024) analyzed more than 11,500 historical wildfire events in Guangdong Province, China, and developed a deep learning model (CNN) for seasonal risk assessment [9]. By incorporating factors such as relative humidity, NDVI index, and temperature, the model achieved high accuracy (AUC = 0.962) and significantly outperformed traditional machine learning methods. In the same year, He et al. proposed an enhanced ConvLSTM deep learning model augmented with attention mechanisms and a Vision Transformer to predict wildfire risk in eastern China [24]. By integrating structural factors such as human activities and land use, the model achieved substantial improvements in predictive accuracy, reaching

an AUC of up to 92.7% and a Kappa coefficient above 0.84. Li et al. (2024) combined deep learning with GIS methods (kernel density analysis, autocorrelation, and centroid shift analysis) to develop a wildfire prediction system for eastern China [73]. Their model, which integrated meteorological, topographic, and socio-economic factors, demonstrated high reliability with an AUC of 88.2%.

In 2025, Liu et al. used Deep Abstract Networks (DANets) alone and in combination with the Whale Optimization Algorithm (WOA) to produce maps of drought and wildfire susceptibility [74]. The samples included 309 fire points (MODIS) and 200 drought points (drought index), and performance was measured by standard metrics (RMSE, MAE, AUC). The results show that the combination of DANets + WOA yields significant improvements—e.g., RMSE and MAE decrease, and AUC increases (from 78.9% to 85.1%), indicating better convergence and reduced overtraining compared to stand-alone models. Xu et al. (2025) conducted a systematic literature review, highlighting that deep learning approaches, particularly CNN, LSTM, and GRU architectures combined with satellite data, consistently outperform traditional machine learning algorithms, such as SVM and Random Forest [6].

On the other hand, Symeonidis et al. (2025) applied ensemble machine learning methods (XGBoost, GBM, LightGBM, and CatBoost) for wildfire susceptibility mapping in Greece [18]. Their models successfully identified up to 83% of wildfires in high-risk areas, underscoring the robustness of ensemble approaches for more stable prediction outcomes.

Although advanced machine learning and deep learning models were used, as well as an extensive database, our study has certain limitations [75–78]. These are reflected in the example of lower-resolution climatological data due to the absence of open data of high-resolution meteorological parameters. This gap in meteorological data could be overcome by installing more local meteorological stations to obtain high-resolution data. The lack of a temporal component of fires is also one of the limitations. Additionally, the temporal discrepancy in the data used represents a certain limitation that can be mitigated by using multiple criteria in the analysis and unifying as much available, open data as possible. However, before integrating with the spatial prediction, it is necessary to analyze in detail the causes of fires in relation to their time of occurrence. On agricultural plots, fires are most often caused by human activity during summer and autumn, whereas in forest areas far from settlements, naturally occurring fires can occur. Such identification of the cause is very complex and challenging, but it would allow a better understanding of the spatio-temporal characteristics of wildfires. When applying spatial and temporal data to predict wildfires, it is very useful to include seasonality, i.e., frequency and intensity, as well as predicting wildfires by season.

While the applied validation strategy balanced spatial and temporal samples, it does not fully capture generalization to new geographic regions or future fire seasons. More advanced schemes, such as spatial cross-validation or reserving the most recent years (e.g., 2023–2024) as an independent blind test, would provide a stricter test of predictive ability. Although this was not feasible in the present study due to data imbalance and computational constraints, it represents an important direction for future research.

5. Conclusions

The frequent occurrence of wildfires in Serbia has significant consequences for the environment and biodiversity. An adequate assessment of the risk of wildfires is crucial for implementing effective preventive measures and mitigating the consequences of disasters. This study presents a spatial prediction of wildfires at the national level, based on 16 quantitative input data (topographic, climatological, hydrological, vegetation, and anthropogenic). The formation of a fire database containing 199,598 fire cases in Serbia enabled the application of deep learning models (KANs and DNN) and the XGBoost algo-

rithm for susceptibility assessment. This study confirms the application and importance of geographic information systems and multisensor remote sensing data (MODIS, VIIRS, Sentinel-2, Landsat 8/9), which have a wide range of applications in predicting natural hazards [79–92].

The results indicate that the DNN model outperforms all other models in all key metrics. It has the highest accuracy of 0.834, the best F1 score of 0.8098, as well as the highest PR-AUC (0.8772) and ROC-AUC (0.9228). This means that it is the most reliable in balancing the recognition of real fires and the avoidance of false alarms, which is especially important for spatial predictions. XGBoost is slightly weaker, but still very good, with an accuracy of 0.8262, an F1 score of 0.8027, a PR-AUC of 0.8708, and a ROC-AUC of 0.9178. KAN is the weakest of the models, with the lowest accuracy (0.8061), the lowest F1 score (0.781), a PR-AUC value of 0.8533, and an ROC-AUC of 0.9008. It can be a useful alternative if a simpler interpretation of the model is important, as it is close in performance to DNN. XGBoost is the weakest of the models, with the lowest accuracy (0.8061), the lowest F1 score (0.781), a PR-AUC value of 0.8533, and an ROC-AUC of 0.9008. Overall, DNN is the most reliable model for fire prediction in this case. KANs remains a strong alternative model, while XGBoost would require further improvement to reach comparable performance.

For the final prediction of fire vulnerability, an ensemble map was generated, which is the product of the three models used. Based on this, 12.5% of the territory in Serbia is highly vulnerable, while 12.7% of the area exhibits very high susceptibility to fire.

An analysis was performed using geographic information systems, specifically examining the overlap between highly and very highly vulnerable areas and administrative units (municipalities) in Serbia. Eight municipalities were identified whose vulnerability exceeds 80% of the total area of the municipality: Žabalj, Žitište, Srbobran, Vrbas, Obilić, Titel, Nova Crnja, and Sečanj.

The SHAP analysis revealed that forest fire risk predictors in Serbia are categorized into several key groups, with varying emphases depending on the model. DNN emphasized the combination of topographical and spatial factors (altitude, forests, agricultural land, settlements, and proximity to water bodies), with the additional contribution of temperature and precipitation, which emphasizes the role of both natural and anthropogenic influences. KANs, on the other hand, emphasized long-term climate aridity and streaks of dry or rainy days, along with elevation and precipitation, indicating that the model is better at recognizing persistent climate patterns than isolated events. XGBoost gave the most weight to direct climate and spatial indicators such as temperature, agricultural land, altitude, and solar radiation, while ranking anthropogenic factors lower. When the results are combined, it becomes clear that altitude, climatic conditions (including temperature, precipitation, aridity, and dry/rainy day sequences), and land use type (such as forests and agricultural land) are the most consistent drivers of fire risk. At the same time, the influence of human activities is more pronounced with deep neural networks than with XGBoost.

This study's results are highly significant and applicable to decision-making in emergency situations. Given that this research encompasses the entire country, the final data will provide a more comprehensive understanding of the spatial characteristics of fires to decision-makers at both national and local levels. A joint approach by state services and local organizations for emergency management and nature protection will enable a more efficient allocation of resources for emergencies. In practice, this means that in areas highly threatened by fires, it would be necessary to carry out more frequent monitoring using patrols, drones, and the installation of fire detection sensors. Such measures should increase the level of nature protection and mitigate the consequences of wildfires.

Future research should integrate the temporal component with the spatial prediction to better understand the timing and location of fire occurrences. Additionally, the appli-

cation of high-resolution climatological data would contribute to obtaining even more precise results, allowing for the clear zoning of the most endangered areas in terms of the environment.

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